Proprietary & Confidential

# From Code to Cloud: Delivering AI-Ready Golden Paths with GKE & Gemini

**Developers write agent logic. The platform makes it safe.**

Sirisha Karra · Cloud Engineer, Google

Google Cloud





# Is an AI agent running in your cluster today?

# Do you know what it's allowed to do or touch?

# From SDLC to the agentic lifecycle

| | Microservices | Agents |
|---|---|---|
| **Behavior** | Deterministic code | Probabilistic reasoning loops |
| **Compute** | Static CPU node pools | Bursty GPUs and TPUs, queued |
| **Identity** | One service, one role | Acts for a user, delegates |
| **Observability** | Latency and error rate | Reasoning traces and tokens |
| **Quality gate** | Tests pass or fail | Continuous evaluation |

# How ungoverned agents fail in production



### Silent loops

No 500 error. Just tokens burning all night.



### Tool sprawl

Unvetted MCP servers become backdoors.



### GPU starvation

Idle agents hold accelerators others need.



### Data leaks

PII slips into prompts and out to models.

Asking every team to solve these alone doesn't scale.

# Shift the hard parts down into the platform

**Developers**
Prompts · reasoning graph · tool logic

**GKE runtime**
gVisor sandboxes · Kueue + DWS for accelerators · quotas

**Platform services**
Agent identity · Model Armor · tool gateway · tracing + evals



**Shift down**

Security, cost and observability move to the platform

# How Shift-Down Happens: 4 Platform Boundaries

**From developer intent to production safety on GKE:**

## 01
### ISOLATE & RUN
**gVisor & Substrate**

- User-space sandbox
- Sub-second idle resume

## 02
### IDENTIFY & GUARD
**Agent Identity**

- Keyless SPIFFE IAM
- Model Armor screening

## 03
### GOVERN
**Agent Gateway**

- Approved MCP catalog
- Loop circuit breakers

## 04
### OBSERVE
**Quality Flywheel**

- OTel GenAI spans
- Continuous eval gating

01 · ISOLATE & RUN

# Sandbox the code, queue the accelerators

- **Agent Sandbox (gVisor):** User-space kernel protects host node
- **Warm pools:** Instant checkout from pre-warmed pool
- **Egress policy:** Blocks metadata server (169.254.169.254)
- **Kueue + DWS:** JIT GPU/TPU queuing, no idle reservations

```yaml
# Reusable blueprint for agent code
apiVersion: extensions.agents.x-k8s.io/v1alpha1
kind: SandboxTemplate
metadata:
  name: python-runtime
spec:
  podTemplate:
    spec:
      runtimeClassName: gvisor
      automountServiceAccountToken: false
      containers:
      - name: runner
        image: python:3.12-slim
---
kind: SandboxWarmPool
spec:
  replicas: 2
  sandboxTemplateRef: { name: python-runtime }
```

⚡ **Agent Substrate 1.0 (GA on GKE · CNCF Sandbox)**

Snapshots idle agents waiting on models or tools. Frees the pod. Resumes in < 0.5s.

02 · IDENTIFY + GUARD

# Verify every caller, Sanitize every prompt, Control with policies

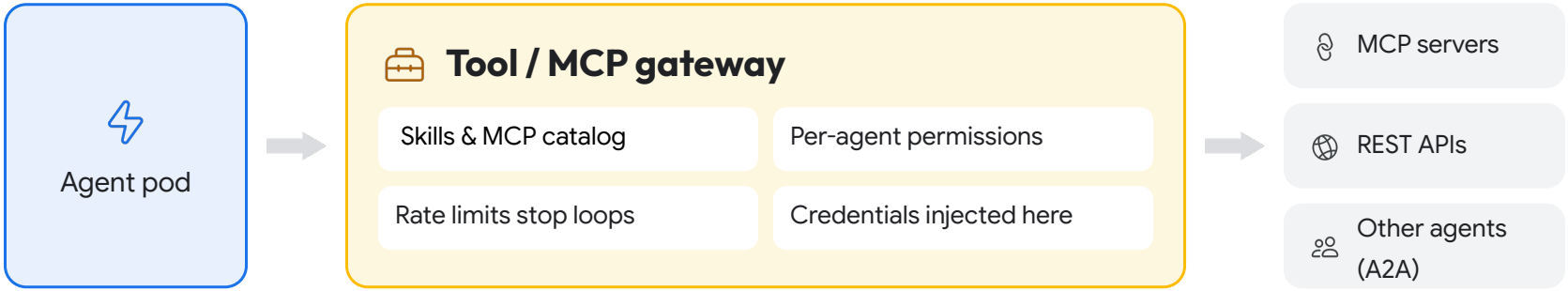
- **User:** signed-in session, passed in context
- **Agent:** dedicated Agent Identity
- **Access Tokens:** short-lived, least-privilege
- Model Armor screens prompts and responses
- Admission policy rejects unsafe agent pods

```
# Grant IAM to the agent's own KSA. No keys.
gcloud projects add-iam-policy-binding $PROJECT \
  --role=roles/bigquery.dataViewer \
  --member="principal://iam.googleapis.com/\
projects/$NUM/locations/global/workloadIdentityPools/\
$PROJECT.svc.id.goog/subject/ns/team-a/sa/invoice-agent"

# Reject agent-code pods that aren't sandboxed
kind: ValidatingAdmissionPolicy
spec:
  validations:
  - expression: >-
      object.spec.runtimeClassName == 'gvisor'
    message: "Agent code must run in gVisor"
```

03 · GOVERN TOOLS

# Every tool call goes through one gateway




```
# Agents may only egress to the gateway
kind: NetworkPolicy
spec: { podSelector: {}, policyTypes: [Egress] }
```

```
egress:
- to: [{ namespaceSelector:
  { matchLabels: { role: tool-gateway } } }]
```

04 · OBSERVE + EVALUATE

# A 200 OK can still be wrong

1

**Trace**

Every step and tool call

2

**Score**

Evaluate live answers

3

**Collect**

Failures become test cases

4

**Gate**

Releases must pass them

```
# OpenTelemetry GenAI span attributes
gen_ai.operation.name: invoke_agent
gen_ai.agent.name: invoice-agent
gen_ai.request.model: gemini-...
gen_ai.usage.input_tokens: 4182
gen_ai.usage.output_tokens: 611
gen_ai.tool.name: bigquery_query

# + k8s.namespace.name, team, user
# → Cloud Trace · Monitoring · BigQuery
```

# The golden path: write 10 lines, get the platform

## DEVELOPER WRITES

```yaml
kind: AgentApp
metadata:
  name: invoice-agent
  team: finance
spec:
  framework: adk
  model: gemini
  tools: [bigquery-readonly, jira-mcp]
  runsGeneratedCode: true
  tokenBudgetPerDay: 2M
```

## PLATFORM INJECTS

- gVisor sandbox + warm pool
- KSA + least-privilege IAM
- Model Armor + admission policy
- Gateway routes + egress policy
- Quota, labels, token budget alert
- OTel tracing + eval pipeline

⚡ **Scaffolded via Antigravity in your IDE**
Uses Google's 'gke-golden-path' skill for production defaults.

🛡️ **Built with Helm, kro, or Crossplane + Backstage**
Platform team defines the boundary; developers ship the logic.

TRY IT LATER

# Build it yourself & References

### Deploy secure AI agents in GKE
ADK agent · Agent Sandbox · Workload Identity · egress policy · warm pools

### Deploy ADK agents to GKE
Package and run an ADK agent on your cluster

### GKE Golden Path Skills & Templates
gke-golden-path skill · AgentApp spec · blueprints



# Thank you

**Questions?**

Find me at the Google Cloud booth ·

**Sirisha Karra**

Connect with me!!

Google Cloud



